**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Variables, Assignment and Data Types

**Name** _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

## Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. What is the difference between input and output?

2. Why is SSD random access often faster than HDD access?

A) Its read head spins faster
B) It uses a laser
C) Its platters are smaller
D) No read head must move

3. "Portability and durability mean exactly the same thing." Correct this. Explain why.

4. A journey planner needs stops and times, but not bus-seat colours. "Abstraction removes every detail, even inputs." Correct this. What should the planner keep?

## Today's objectives

Declare, assign and update variables of appropriate types. Use variable and assignment in a practical scenario.

## Key vocabulary - Read, cover, recall

**Variable**
A variable stores a named value that can change while the program runs.

**Assignment**
The equals sign stores the new value in a variable, replacing its old value.

**Integer**
A whole-number data type is suitable for counts, ages and scores.

**String**
A string stores text such as names or class codes.

**Write one key term and its meaning in your own words:**

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Learn and practise

## New learning - Key ideas

| | |
|---|---|
| **What is Variable?** | Variable - A variable stores a named value that can change while the program runs. |
| **In this example** | A library stock record uses assignment; The equals sign stores the new value in a variable, replacing its old value. |
| **Follow the effect** | A whole-number data type is suitable for counts, ages and scores. Explain how this changes the eventual program result. |
| **Check understanding** | String is used for a specific reason: A string stores text such as names or class codes. |

## I Do - One worked example

**A library stock record is being written. Explain how variable and assignment are used to make the program work. [4 marks]**

1. We are designing a library stock record. Identify which data and decisions are essential.

2. Variable [1] - A variable stores a named value that can change while the program runs. [1].

3. Assignment [1] - The equals sign stores the new value in a variable, replacing its old value. [1].

4. Two linked points for each technique give four mark points.

## We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. In a pupil merit totals, explain how variable helps the program. [2 marks]

2. For a sports-club members, explain why assignment matters. [2 marks]

3. When designing a cinema seats remaining, explain how integer and string should be used. [4 marks]

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Think and write

## Code First - Predict > Write > Test > Improve

**YOUR TASK: Use the named constant to update the free RAM.**

```
const RESERVED_MB = 120
freeMB = 500
freeMB = ...
print(freeMB)
```

BLACK PEN: predict or complete the code; test, then improve.

## You Do - Independent exam questions

Q2. State two programming terms or techniques relevant to this lesson. [2 marks]

Q3. A school register needs to work correctly. Explain how assignment and integer would help. [4 marks]

Q4. A pupil is programming a shop loyalty points. Explain the roles of variable, integer and string in their solution. [6 marks]

## Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. A variable stores a named value that can change while the program runs. ☐ **T** ☐ **F**
2. Variable - A variable stores a named value that can change while the program runs. ☐ **T** ☐ **F**
3. A library stock record uses assignment; The equals sign stores the new value in a variable, replacing it... ☐ **T** ☐ **F**

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

## Do Now - Answers

**Q1** Input enters a program; output is information the program produces. ☐

**Q2** D - No read head must move ☐

**Q3** No moving read/write parts makes an SSD less likely to be damaged by shocks. Size and weight also affect portability. ☐

**Q4** Remove details that are not needed. Keep the information and parts needed to solve the problem, then link clear steps. ☐

## Code First - Model

```
const RESERVED_MB = 120
freeMB = 500
freeMB = freeMB - RESERVED_MB
print(freeMB)
```

**Output: 380**

## We Do - Check the model

1. Variable [1] - A variable stores a named value that can change while the program runs. [1]. ☐

2. Assignment [1] - The equals sign stores the new value in a variable, replacing its old value. [1]. ☐

3. Integer [1] - A whole-number data type is suitable for counts, ages and scores. [1]. String [1] - A string stores text such as names or class codes. [1]. ☐

## Exit Ticket - Check

**1. TRUE 2. TRUE 3. TRUE**

## You Do - Self-mark

**Q2** Variable [1]; Integer [1]. ☐

**Q3** Assignment [1] - The equals sign stores the new value in a variable, replacing its old value. [1]. Integer [1] - A whole-number data type is suitable for counts, ages and scores. [1]. ☐

**Q4** Variable [1] - A variable stores a named value that can change while the program runs. [1]. Integer [1] - A whole-number data type is suitable for counts, ages and scores. [1]. String [1] - A string stores text such as names or class codes. [1]. ☐

## Try the question again

**1. Copy the FULL question you found most difficult:**

**2. In GREEN PEN, write a complete new answer:**